MIL-C-45526B(WC)
19 July 1972
SUPERSEDING
MIL-C-45526A(WC)
15 October 1968
AMENDMENT 1
20 November 1968

MILITARY SPECIFICATION

CANNON, 105MM HOWITZER: M103

1. SCOPE

1.1 This specification covers one type of howitzer cannon.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-C-13931 - Cannons, General Specification for
MIL-I-45607 - Inspection Equipment Acquisition, Maintenance, and Disposition of

STANDARDS

Military

MIL-STD-109 - Quality Assurance Terms and Definitions
MIL-STD-1235 - Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes

PUBLICATIONS

U. S. Army Weapons Command

EPL 8768965 - Engineering Parts List for Cannon, 105MM Howitzer: M103
Master LSQAP 8768965 - Master List of Supplementary Quality Assurance Provisions for Cannon, 105MM Howitzer: M103
DL 8768965 - List of Inspection Equipment for Cannon, 105MM Howitzer: M103
PDS P8768965 - Packaging Data Sheet, Cannon, 105MM Howitzer: M103

(Copies of specifications, Standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

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3. REQUIREMENTS

3.1 First article. When specified in the contract, the contractor shall submit a first article. Unless otherwise specified (see 6.1.1), the first article shall be included in the pilot pack (see 5.1).

3.2 General requirements. Cannons and parts thereof shall conform to the drawings listed in Engineering Parts List 8768965 and the requirements of this specification and MIL-C-13931.

3.3 Functioning. All parts shall function smoothly without interference or erratic movement.

3.3.1 Breech mechanism. Camming of the operating-crank during counterrecoil or rearward rotation of the operating-handle shall rotate the breechblock crank rearward and downward to lower the breechblock. When the breechblock nears the opened position, the lips of the extractors shall rotate sharply rearward to extract the cartridge case. The extractors shall then lock the breechblock in the opened position until tripped, normally by insertion of a cartridge. The breech mechanism, with the closing-spring-adjustor set for minimum preload, shall then raise the breechblock to the closed position. Counterclockwise rotation of the adjustor shall increase the preload of the closing spring.

3.3.2 Firing mechanism group. During the dropping of the breechblock, the cocking lever, forced by a camming surface on the breech ring, shall engage the retractor. This engagement shall force the percussion mechanism rearward. The firing spring shall then compress and cause the sear to latch the percussion mechanism in the cocked position. Rearward movement of the trigger plunger pivots the trigger. The sear shall then rotate and release the percussion mechanism. This releases the compressed firing spring which shall drive the firing pin forward. The firing pin shall strike the cartridge primer and simultaneously compress the firing pin retracting spring. The expanding retracting spring shall then withdraw the firing pin behind the front face of the breechblock.

3.3.3 Firing pin protrusion. The firing pin protrusion, measured from the front face of the breechblock, shall be not less than 0.10 inch nor more than 0.16 inch when the breechblock is in the firing position.

3.3.4 Firing mechanism overtravel. Overtravel of the firing mechanism in cocking shall be not less than 0.05 inch.

3.3.5 Headspace. Headspace, the distance between the front face of the closed and locked breechblock and the rear face of the tube, shall be not less than 0.188 inch nor more than 0.206 inch.

3.4 Performance.

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3.4.1 High pressure resistance. The cannons and repair tubes shall be capable of withstanding 113 ± 4 percent of the 39,000 pounds per square inch upper pressure limit at 70°F when tested as specified in 4.5.4.

3.4.2 Material soundness. After testing, all parts shall be free from cracks and fractures and shall comply with any soundness requirements specified on the applicable drawings.

3.4.3 Bore enlargement. If proof firing is used to evaluate high pressure resistance, bore diameters, measured across the lands after proof firing no more than 8 rounds, shall not exceed the basic (minimum) diameter specified on the applicable drawing by more than the amount specified in Table I.

Table I Bore enlargement

Distance forward of commencement of rifling	0.1 inch	1 inch	1 cal.	2 cals. to muzzle
Amount (inches)	0.008	0.008	0.006	0.003

NOTE: Interpolate to determine the amount between given distances.

3.5 Workmanship. Workmanship shall be as required in MIL-C-13931.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Quality assurance terms and definitions. Quality assurance terms and definitions shall be as specified in MIL-STD-109.

4.3 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.4)
b. Quality conformance inspection (see 4.5)

4.4 First article inspection. A first article shall be submitted for inspection (in accordance with the contract requirements). The first article shall be representative of the production processes to be used during quantity production. The first article shall be subjected to the quality conformance inspection specified herein and such other inspection as is necessary to determine compliance with the requirements of the contract.

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4.5 Quality conformance inspection.

4.5.1 Inspection provisions. Cannon shall be inspected for conformance to requirements of the provisions of this specification, MIL-C-13931, SQAPs listed in Master LSQAP 8768965, and SQAP-APPENDIX-WVT.

4.5.1.1 Final examination. Final examination of each cannon shall be completed after testing and prior to preservation and packaging to determine compliance with this specification, MIL-C-13931, and as specified in the contract.

4.5.1.2 Inspection approval stamp. The application of the inspection approval stamp shall be as specified in MIL-C-13931.

4.5.2 Inspection equipment. Data lists containing equipment used in the inspection of parts, sub-assemblies, and assemblies are listed in DL 8768965.

4.5.2.1 Acquisition, maintenance, and disposition. Unless otherwise specified, responsibility for acquisition, maintenance, and disposition of acceptance inspection and test equipment prescribed in the List of Inspection Equipment DL 8768965, and for all other inspection equipment required to perform inspection prescribed by applicable specifications, shall be in accordance with MIL-I-45607.

4.5.2.2 Accuracy of standard measuring equipment. When commercial and modified commercial inspection and test equipment is used, it shall be capable of repetitive measurements to an accuracy of 10 percent of the total tolerance of the characteristic being inspected.

4.5.3 Examinations.

4.5.3.1 Firing pin protrusion. Firing pin protrusion shall be gaged on each cannon with Gage 7248891 or an approved equivalent gage. Retraction shall be checked visually. Cannons failing to meet the requirements of 3.3.3 shall be rejected.

4.5.3.2 Firing mechanism overtravel. Firing mechanism overtravel shall be gaged on each breechblock before insertion in the breech ring. Overtravel shall be measured with Gage 8773831 or an approved equivalent gage. Procedure shall be as follows: Remove the firing spring retainer, insert the gage and hand cock the mechanism until the sear locks the firing mechanism. Allow the cocking lever to return to its forward position. Set the dial indicator of the gage to zero. Pull the cocking lever rearward until the lever cam surface is in line with the frontface of the breechblock guide surfaces. The indicator reading at this position is the measurement of minimum overtravel. Cannons failing to meet the requirements of 3.3.4 shall be rejected.

4.5.3.3 Headspace. Headspace shall be gaged on each cannon with Gages 8773818 and 8773819 or with approved equivalent gages. Cannons failing to meet the requirements of 3.3.5 shall be rejected.

4.5.3.4 Workmanship. Workmanship shall be examined in accordance with MIL-C-13931.

4.5.3.5 Preparation for delivery. The classification of defects and the AQLs for examination of the packaging, packing, and marking are as follows:

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Category	Defect	AQL	Insp. Method
Major			
101	Illegible or incorrect marking	1.0	Visual
102	Improper level of preservation, packaging, or packing	1.0	Visual
103	Inadequate cleaning or drying	1.5	Visual
104	Improper preservative application	1.5	Visual
105	Missing or improper cushioning	1.5	Visual
106	Improper blocking or bracing	1.5	Visual/Manual
107	Improper closure or strapping of container	1.5	Visual/Manual
Minor			
201	Workmanship	4.0	Visual

4.5.4 Tests.

4.5.4.1 Functioning. Each cannon shall be operated manually to test smoothness of operation. Cocking action, trigger actuation, firing mechanism action, and breech mechanism operation tests shall be performed to ascertain proper functioning. Extraction shall be tested with an empty service cartridge case and with the cannon in a horizontal position. Cannons failing to meet the requirements shall be rejected.

4.5.4.2 Performance.

4.5.4.2.1 Actual firing. Unless otherwise specified in the contract each cannon or repair tube shall be subject to actual firing by Government personnel at a proving ground designated by TECOM. Unless otherwise specified in the contract, the cannon or repair tube firing schedule shall consist of firing one round of ammunition which produces 113% ± 4% of the UPL (Upper Pressure Limit). Unless otherwise specified in the contract, sampling for repair tubes shall be at an AQL of 0.10% in accordance with MIL-STD-1235. (see 6.1.1).

4.5.4.2.2 Simulated firing. When specified in the contract (see 6.1.1), breech mechanisms shall be subjected to simulated firing in accordance with procedures specified in the contract.

4.5.4.2.3 Material soundness. After firing, each cannon or repair tube shall be examined visually for freedom from cracks, fractures, and other defects. Furthermore, the breechblock, breechring, evacuator chamber, and the outer surfaces of the tube shall be magnetic particle tested in accordance with the applicable part drawing and specification. The inner surfaces of the tube shall be examined by means of a borescope. Any evidence of cracks or fractures shall be cause for rejection.

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4.5.4.2.4 Bore enlargement. After firing no more than 8 rounds, the bore diameter across the lands of each cannon or repair tube shall be measured with an air gage, a star gage or other approved gage. Measurements shall be recorded. Enlargements exceeding the limits specified in Table I shall be cause for rejection.

4.5.4.2.5 Records. Cannons or repair tubes that pass the firing tests shall be furnished with applicable data on DA Form 2408-4 by the proving ground. Repair tubes accepted on a proof sampling basis shall be furnished with applicable data on DA Form 2408-4 by the contractor and approved by the Government representative. The responsibility of the contractor for cannons or repair tubes which fail the firing tests shall be as specified in the contract.

5. PREPARATION FOR DELIVERY

5.1 Pilot pack. On each contract, a pilot pack shall be forwarded in accordance with 3.1. Pilot packs for the cannon shall be packaged to the level of packaging specified in the contract and packed level C in accordance with the requirements of Packaging Data Sheet P8768965.

5.2 Preservation, packaging, and marking. The cannon shall be unit packaged, packed, and marked in accordance with the requirements of Packaging Data Sheet P8768965 for the level of protection specified in the contract (see 6.1).

5.3 Repair parts. Repair parts shall be unit packaged, packed and marked in accordance with the requirements of the applicable packaging data sheet (see 6.1.1).

6 NOTES

6.1 Ordering data. Procurement documents should specify the following:

6.1.1 Procurement requirements.

a. Title, number, and date of this specification.

b. Requirements for submission of and shipping instructions for first article (see 3.1 and 4.4) and pilot pack, if different (see 3.1 and 5.1).

(1) The contract should designate the examinations and tests to be performed by the contractor and the examinations and tests to be performed by the Government.

(2) The contract should specify the sample size (i.e., the number of units) of the first article.

(3) The contract should designate the inspection and test data that is required to be furnished the Government when the contractor is required to perform either or both examinations and tests.

(4) The contract should designate at whose expense a retest may be performed.

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(5) The contract should specify that the acceptable first article sample shall be delivered in accordance with the terms of the contract, or

(6) Any additional or extended examinations and tests beyond the scope of this specification shall be specified and identified as a separate item of the contract.

c. Serial numbers for the cannons and tube assemblies.

d. List of inspection equipment, responsibility for acquisition, maintenance, and disposition thereof, if other than as specified (see 4.5.2).

e. Extent of supplier's responsibility for Government-furnished equipment.

f. Selection of applicable level of preservation, packaging, and packing (see 5.2).

g. Packaging instructions for repair parts (see 5.3).

h. The contract should specify under what conditions simulated firing may be performed, the test equipment available, and the procedures to be followed.

i. When applicable, the contract should specify the sampling plan for the actual firing of repair tubes.

j. The contract should specify the responsibility of the contractor for material which fails proof firing tests.

6.1.2 Contract data requirements. Monthly reports of the results of final examination and performance testing shall be specified for delivery on DD Form 1423 in the contract.

6.2 When warranted, the contract should specify the application of MIL-Q-9858 or MIL-I-45208, as appropriate, on the Management Control Systems Summary List, DD Form 1660.

6.3 Unless otherwise specified, the contract should specify the application of MIL-I-45607 and MIL-C-45662 on the Management Control Systems Summary List, DD Form 1660.

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426

REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED.

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